

On the learning-based control of continuum robots with provable robustness, efficiency, and generalizability

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Abstract

Recent years have witnessed the remarkable advancements in Koopman-operator-based data-driven methods for continuum robot control. However, there is currently a paucity of both theoretical and practical work investigating the convergence and robustness of these methods, which is crucial due to the complexity and susceptibility of continuum robots and the training-to-reality gap. This work seeks to complete Koopman-operator-based methods in terms of accuracy, robustness, generalizability, and theoretical analysis while maintaining high computational efficiency. In this work, we learn the continuous-time model of continuum robots using a deep Koopman network, which bridges the gap between unknown robot models and model-dependent iterative learning control, and propose a novel control framework for data-driven control of continuum robots. Rigorous theoretical analysis is then provided to prove the convergence and robustness of the proposed method. Finally, comprehensive comparisons of three types of Koopman-operator-based methods are conducted, using six metrics to evaluate their performance. Experiments on two heterogeneous continuum robots indicate that our proposed method outperforms existing Koopman-operator-based control methods across most metrics, significantly improving robustness and generalizability. Furthermore, this work has guiding significance for applying Koopman-operator-based control methods to the efficient and robust control of other robotic systems.

Keywords

Continuum robots, data-driven control, provable robustness

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1. Introduction

Continuum robots have received increasing attention in recent decades. Their compliance and dexterity make them valuable in various important areas such as medical treatment (Burgner-Kahrs et al., 2015; Li et al., 2024), industry (Dong et al., 2019), agriculture (Shao et al., 2015; Yeshmukhametov et al., 2022), and aerospace (Dong et al., 2022; Peng et al., 2023). Fully exploiting their capabilities necessitates the design of effective, efficient, and robust control systems for continuum robots, which remains challenging due to their continuous nature.

Numerous attempts have been made to establish analytical models for continuum robots and perform control based on these models (Della Santina et al., 2019; Shah et al., 2021; Rao et al., 2023; Wang et al., 2024; Mbakop et al., 2024; Ferguson et al., 2024; Zhang et al., 2025). Representative approaches for modeling continuum robots include the piecewise constant curvature model (Della Santina et al., 2020a, 2020b), the finite element method (Goury and Duriez, 2018), the Cosserat rod model (Lilge

and Burgner-Kahrs, 2023; Tummers et al., 2024), and the Euler–Bernoulli beam theory (Sadati et al., 2021). Based on these established models, control of continuum robots can be achieved using closed-form analytical inverse kinematics solutions (Neppalli et al., 2009), solutions computed from the robot Jacobian (Santoso and Onal, 2021), or solutions obtained through iterative searching (Kolpashchikov et al., 2018). Although model-based control of continuum robots has made significant progress, it was argued that analytical modeling of continuum robots is challenging because their configurations are rarely sensed completely and are easily

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affected by environmental interactions (Yip and Camarillo, 2016), which hinders the application of model-based approaches.

Due to the complexity and difficulty of analytical modeling, numerous data-driven control methods have been proposed to estimate or learn the kinematics, inverse kinematics, or differential kinematics of continuum robots (Chen et al., 2025; Hu et al., 2023; Nuelle et al., 2020; Yu et al., 2024; Zhang et al., 2024). For instance, Tan et al. (2023, 2025) proposed estimating the kinematic Jacobian using sensory feedback data and controlling continuum robots through Jacobian inversion. However, accurately estimating the kinematic Jacobian is challenging, especially when the degrees of freedom (DoF) in the actuation space are redundant. Thuruthel et al. (2017) reported a neural-network-based method to directly learn the inverse kinematics of continuum robots. Kasaei et al. (2023) developed a data-efficient neural network to approximate the differential kinematic model and estimate the robot Jacobian for control purposes. However, Bruder et al. (2019) argued that neural-network-based methods only provide a black box since no explicit model is constructed, rendering these methods incompatible with existing model-based control techniques. To address this problem, they proposed learning the robot model in a lifted space using Koopman operator and controlling robots with the model predictive control (MPC) method. Likewise, Haggerty et al. (2023) introduced a Koopman-operator-based linear quadratic regulator (KLQR) method to control continuum robots with inertial dynamics. Since the effectiveness of the Koopman operator relies on selecting appropriate lifting functions, and manually choosing these functions is challenging, some works have proposed learning lifting functions and the discrete-time Koopman operator simultaneously using deep neural networks (Chen et al., 2024; Lusch et al., 2018; Shi and Meng, 2022). Given their superior performance, Koopman-operator-based control methods have garnered significant interest (Bruder et al., 2021; Thamo et al., 2023; Wang et al., 2023). However, these methods still suffer from some limitations. Firstly, MPC is well-known for its computational expense, which may affect the real-time performance of Koopman-operator-based MPC (KMPC) methods under high sampling frequencies. Although the KLQR method is computationally efficient, neither KMPC methods nor KLQR methods can adequately compensate for uncertainty and disturbance, resulting in subpar performance due to the training-to-reality gap illustrated in Figure 1(a). This gap can easily degrade the task-space performance of continuum robots in real-world scenarios due to their inherent susceptibility. Additionally, as shown in Figure 1(b), a robot may need to operate in areas not covered during training, and the structural diversity of continuum robots further complicates control. Moreover, the convergence and robustness of existing Koopman-operator-based methods have not been studied either theoretically or experimentally. Thus, developing a data-driven control algorithm with significantly improved robustness, high computational

efficiency, strong generalizability, and rigorous theoretical analysis is essential for continuum robots.

Adaptive iterative learning control (ILC), an effective intelligent control method for solving repetitive tracking control problems, has been extensively studied due to its computational efficiency and its capability to handle uncertainties and disturbances (Li and Huang, 2024). However, several challenges remain in applying ILC to continuum robots. Many ILC methods focus on single-input-single-output (SISO) or square multiple-input-multiple-output (MIMO) systems, whereas a lot of continuum robots are classified as non-square MIMO systems. Additionally, although some studies address non-square MIMO systems (Xu and Tan, 2002), they assume the input distribution matrix is left-invertible, which is not always the case. To address these challenges, Li and Hou (2024) proposed an inversion-free adaptive ILC method for non-square nonlinear systems. However, this method requires the access to the input distribution matrix, which must be computed based on the analytical robot model when applying the control method to continuum robots. The modeling complexity and difficulty of continuum robots may render these model-dependent ILC methods less effective and less generalizable across different robots.

This work aims to propose a control algorithm that demonstrates good generalizability across different types of continuum robots, such as the heterogeneous continuum robots shown in Figure 1. On one hand, the structural complexity of continuum robots poses significant challenges for analytical modeling. On the other hand, a unified equation describing continuum robot models remains an open research topic currently being investigated by different communities (Della Santina et al., 2023). If a control algorithm is designed relying on the robot model or parts of it, it is foreseeable that the generalizability of such an algorithm would be significantly reduced across heterogeneous robots. Therefore, one of the key objectives of this work is to achieve the control of continuum robots in situations where the robot model is completely unknown. Data-driven approaches are promising to achieve this goal, especially those based on the Koopman operator, which can efficiently learn the model of the robot from data, enabling effective control.

However, existing Koopman-operator-based control methods for continuum robots cannot compensate for uncertainties and disturbances, and their convergence and robustness lack theoretical guarantees. ILC has the potential to address these issues. Although ILC is highly robust to model uncertainties, it still requires partial knowledge of the system model. For instance, Pierallini et al. (2023b) implemented iterative control of continuum robots under the assumption that model knowledge is available. Although some methods can achieve ILC in cases where the system model is entirely unknown (Xu et al., 2014), this is typically only applicable to square systems (i.e., systems where the state and input dimensions are the same). In non-square systems, which are common in continuum robots, partial

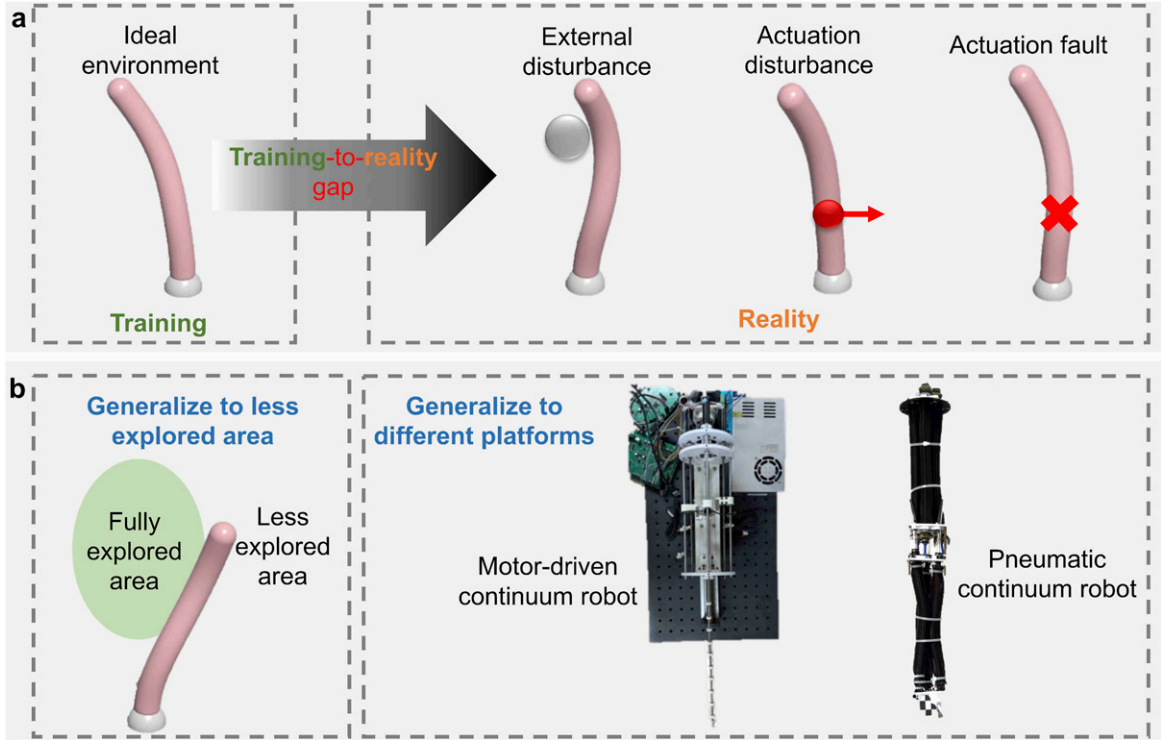


Figure 1. Robustness and generalizability in continuum robot control. (a) A well-trained control algorithm for continuum robots may encounter unexpected situations in reality such as external disturbance, actuation disturbance, and actuation fault. Therefore, developing a robust control algorithm that can bridge the training-to-reality gap is crucial for continuum robots. (b) A robust control algorithm is also expected to generalize well to less-explored area in the training stage and to platforms with different mechanical structures, given the diversity of continuum robots.

knowledge of the system model is still required (Li and Hou, 2024; Xu et al., 2014). Therefore, under the premise considered in this paper (i.e., the robot system model, which could be non-square, is completely unknown), these existing ILC algorithms cannot be directly applied to continuum robots. The Koopman operator is promising to bridge this gap, as it can learn an explicit model of the robot from data.

The complementary nature of Koopman modeling and ILC motivates us to combine the two approaches to develop an efficient and robust data-driven ILC method for continuum robots. ILC algorithms are commonly used in the discrete-time domain (Bristow et al., 2006), and the Koopman operator is often employed to learn discrete-time models of continuum robots. Therefore, the combination of these two methods is straightforward and natural. However, it is not enough to simply combine them. Continuum robots inherently exhibit continuous-time dynamics and many control problems are naturally expressed in continuous time (Della Santina and Angelini, 2021), while existing Koopman-operator-based methods typically construct discrete-time models with fixed time steps for continuum robots, leading to decreased prediction performance in scenarios that require varying or different time steps. Besides, discrete models with fixed time steps further impose limitations on the controller's operating frequency. Addressing these issues often necessitates re-collecting data and retraining the model,

which is non-trivial. Therefore, it is essential to leverage the Koopman operator to learn the continuous-time model of the robot, which is independent of the sampling interval of time, and design a corresponding ILC algorithm.

Although Saltik et al. (2022) have preliminarily explored combining the Koopman operator and ILC for controlling repetitive nonlinear systems, the ILC signal was computed offline, and the online control signal was still computed using MPC, which also suffers from the limitations of poor efficiency and discrete-time dynamics. Therefore, as illustrated in Figure 2, the primary contributions of this work are summarized as follows:

- (1) We present, for the first time, a method for learning the model of continuous-time dynamic systems using a deep Koopman network (DKN). By utilizing input-output data from the original nonlinear system, the DKN is trained to simultaneously learn the lifting function and the continuous-time linear evolution model in the lifted space.
- (2) By integrating the learned model with ILC, we propose a novel control framework for data-driven tracking control of continuum robots. This marks the first proposal of a Koopman-operator-based ILC (KILC) method for continuum robots.
- (3) Through rigorous theoretical analysis, we prove that the proposed KILC method guarantees the

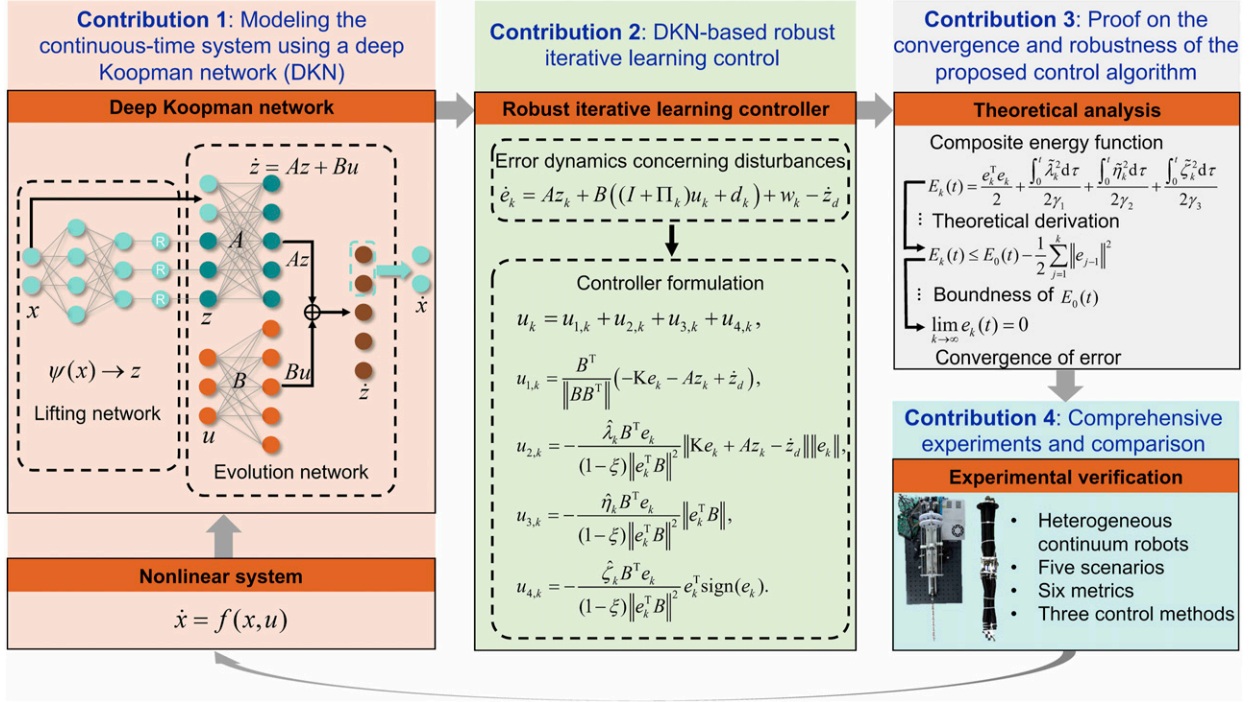


Figure 2. Illustration of contributions. This work makes the following four contributions: (1) learning the continuous-time nonlinear model of continuum robots using a deep Koopman network; (2) proposing a DKN-based iterative learning controller; (3) providing rigorous convergence and robustness analysis for the proposed control algorithm; (4) verifying the proposed algorithm on heterogeneous continuum robots in various scenarios.

convergence of trajectory tracking errors as the iteration number increases, even in the presence of various uncertainties and disturbances. This represents the first theoretical analysis of the convergence and robustness of Koopman-operator-based control algorithms for continuum robots.

- (4) By conducting comprehensive experiments on two heterogeneous continuum robots, we evaluate and compare the performance of different Koopman-operator-based control methods for continuum robots in terms of accuracy, robustness, generalizability, and computational efficiency.

2. Preliminary

This section elaborates some preliminary knowledge to facilitate the understanding of the problem and approaches.

2.1. Problem statement

This work focuses on the control problem of continuum robots, which can be characterized by the following nonlinear dynamic system:

$$\begin{aligned} \dot{s}(t) &= f(s(t), u(t)), \\ x(t) &= s(t) \end{aligned} \quad (1)$$

where $s(t) \in \mathbb{R}^m$ is the m -dimensional system state at the time instant t , $x(t) \in \mathbb{R}^n$ is the system output, $u(t) \in \mathbb{R}^n$

represents the n -dimensional system input, $\dot{s}(t)$ is the first time derivative of $s(t)$, and $f: \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ denotes a nonlinear mapping function determined by the system model and parameters. It is worth noting that the system can be non-square, meaning that $m \neq n$. This work primarily focuses on common continuum robot systems, where the dimension of the input space is typically greater than or equal to that of the state space, that is, $m \leq n$. This ensures that the robot can complete specific tasks, such as trajectory tracking. However, when $m > n$, the robot becomes an underactuated system, in which case it is difficult to guarantee task completion. Such scenarios are not considered within the scope of this work.

Specifically, this work seeks to solve the end-effector tracking control problem of continuum robots. Therefore, the end-effector position is taken as the system output $x(t)$ of (1). Our objective is to develop an iterative learning control system, which can generate appropriate input signal $u_k(t)$ in the k^{th} iteration, such that the high-precision tracking performance can be achieved in the iterative domain, which is formulated as

$$\lim_{k \rightarrow \infty} (x_k(t) - x_d(t)) = 0, \forall t \in [0, T_d] \quad (2)$$

where $x_k(t)$ denotes the actual end-effector trajectory in the k^{th} iteration, $x_d(t) \in \mathbb{R}^n$ denotes the reference trajectory, and T_d is the task duration.

Unfortunately, unlike simulative environments (Naughton et al., 2021), the system model f is often

unknown in reality and only measurements of the system output are available when it comes to continuum robots (Lusch et al., 2018), which poses challenges to their control.

2.2. Iterative learning control

Since the control method in this work is inspired by the adaptive iterative learning control (AILC) method proposed by Li and Huang (2024), we briefly introduce the key steps of the AILC method below. The argument t will be dropped if no confusion is caused hereinafter.

For the MIMO nonlinear system considered by the AILC method, the corresponding error dynamics is

$$\dot{\varepsilon}_k = f(x_k)\Theta + \mathcal{B}((I + \Pi_k)u_k + d_k) + w_k \quad (3)$$

where $\varepsilon_k = x_k - x_d$ is the error between the actual system output and reference output, $f(x_k)$ denotes a known function, Θ is the unknown parametric uncertainty, $\mathcal{B}$ is a known non-square control input distribution matrix, I is the identity matrix, Π_k denotes the input distribution uncertainty, and d_k and w_k are the state-dependent matched and mismatched disturbances, respectively.

To compensate for uncertainties while achieving perfect tracking performance, the AILC method is formulated as

$$u_k = \begin{cases} u_{1,k} + u_{2,k} + u_{3,k} + u_{4,k}, & \|\varepsilon_k^T \mathcal{B}\| \neq 0 \\ u_{1,k}, & \|\varepsilon_k^T \mathcal{B}\| = 0 \end{cases} \quad (4)$$

where

$$u_{1,k} = \frac{\mathcal{B}^T}{\|\mathcal{B}\mathcal{B}^T\|} \left(-K\varepsilon_k - f(x_k)\hat{\Theta} \right), \quad (5)$$

$$u_{2,k} = -\frac{\hat{\lambda}_k \mathcal{B}^T \varepsilon_k}{(1 - \xi) \|\varepsilon_k^T \mathcal{B}\|^2} \|K\varepsilon_k + f(x_k)\hat{\Theta}\| \|\varepsilon_k\|, \quad (6)$$

$$u_{3,k} = -\frac{\hat{\eta}_k \mathcal{B}^T \varepsilon_k}{(1 - \xi) \|\varepsilon_k^T \mathcal{B}\|^2} \|\varepsilon_k^T \mathcal{B}\|, \quad (7)$$

$$u_{4,k} = -\frac{\hat{\zeta}_k \mathcal{B}^T \varepsilon_k}{(1 - \xi) \|\varepsilon_k^T \mathcal{B}\|^2} \varepsilon_k^T \text{sign}(\varepsilon_k), \quad (8)$$

with K being a positive-definite feedback gain matrix, $\xi \in (0, 1)$ being a tunable parameter, and $\hat{\lambda}_k$, $\hat{\eta}_k$, and $\hat{\zeta}_k$ being the estimations of some unknown parameters. Besides, $\hat{\Theta}$ is the estimation of Θ generated by

$$\hat{\Theta}_k = \hat{\Theta}_{k-1} + \Gamma f^T(x_k) \varepsilon_k, \quad \hat{\Theta}_{-1} = 0 \quad (9)$$

where Γ is a positive-definite gain matrix. To ensure convergence, the following parameter updating laws are utilized:

$$\hat{\lambda}_k = \begin{cases} 0, & k = -1 \\ \hat{\lambda}_{k-1} + \gamma_1 \|f(x_k)\hat{\Theta}\| \|\varepsilon_k\|, & k \geq 0 \end{cases}, \quad (10)$$

$$\hat{\eta}_k = \begin{cases} 0, & k = -1 \\ \hat{\eta}_{k-1} + \gamma_2 \|\varepsilon_k^T \mathcal{B}\|, & k \geq 0 \end{cases}, \quad (11)$$

$$\hat{\zeta}_k = \begin{cases} 0, & k = -1 \\ \hat{\zeta}_{k-1} + \gamma_3 \varepsilon_k^T \text{sign}(\varepsilon_k), & k \geq 0 \end{cases}, \quad (12)$$

where γ_1 , γ_2 , and γ_3 are learning rates to be designed.

Although the AILC method has shown excellent performance in the control of nonlinear systems with various uncertainties, it requires the availability of the function $f(x_k)$ and input distribution matrix $\mathcal{B}$, which need to be derived from the analytical model of the system. However, due to the complexity and difficulty in analytical modeling of continuum robots, deriving $f(x_k)$ and $\mathcal{B}$ from the mapping function f becomes non-trivial. Therefore, it is challenging to directly apply the AILC method to the continuum robot described by (1).

2.3. Modeling based on Koopman operator

To avoid using analytical models, Koopman operator theory (Koopman, 1931) has been adopted to learn the unknown dynamics in a data-driven manner. Koopman operator provides an explicit and linear model for continuum robots and facilitates the application of model-based control methods. The basic idea is that any finite-dimensional, Lipschitz continuous nonlinear dynamic system has an equivalent infinite-dimensional linear representation in the space of all real-valued functions of the system state (Bruder et al., 2019). Accordingly, it was reported that the nonlinear dynamic system (1) is approximately equivalent to the following linear dynamic system in a high-dimensional space:

$$\begin{aligned} z(t_{i+1}) &= Az(t_i) + Bu, \\ x &= Cz \end{aligned} \quad (13)$$

where $z(t_i)$ denotes the state vector in the high-dimensional space at the i^{th} discrete time instant, which is obtained by lifting the robot state x in the original space, and A , B , and C are constant matrices learned from data or designed manually.

Despite the superior performance of Koopman-operator-based modeling methods for continuum robots (Bruder et al., 2019; Haggerty et al., 2023; Thamo et al., 2023), there are still some disadvantages of them. First, there is a lack of effective and efficient method for selecting lifting functions. Although it was reported that the lifting functions and system matrices can be learned simultaneously using deep neural networks (Chen et al., 2024; Lusch et al., 2018; Shi and Meng, 2022), the validation on the control of continuum robots is rarely reported. Second, physical systems, such as the continuum robot described by (1), inherently exhibit continuous-time dynamics. However, existing Koopman-based methods typically construct discrete-time models for continuum robots. These discrete-time models are typically constructed with a fixed time step,

leading to decreased prediction performance in scenarios that require varying or different time steps. Addressing such issues often necessitates re-collecting data and retraining the model. Furthermore, discrete models with fixed time steps impose limitations on the controller's operating frequency. Therefore, a method that can learn continuous-time models is indispensable.

3. Method

In this section, we formulate the modeling and control methods proposed in this work and provide the theoretical analysis regarding the convergence and robustness of the proposed control method.

3.1. Modeling continuous-time system using a deep Koopman network

In order to address the problems faced by conventional discrete-time Koopman-operator-based methods, the first contribution of this work is to learn a continuous-time model of continuum robots described by (1) using a deep Koopman network.

In the infinite-dimensional observable space $\mathcal{H}$ (referred to as the lifted space), a Koopman operator $\mathcal{K}$ can describe the evolution of functions of the system state. Specifically, the mathematical formulation of this evolution is given by

$$\frac{d}{dt}\psi(x, u) = \mathcal{K}\psi(x, u) \quad (14)$$

where $\psi(x, u)$ is observable (i.e., the lifting function). Since the Koopman operator is infinite-dimensional, it is preferable to identify a finite-dimensional approximation of the operator via the extended dynamic model decomposition algorithm (Williams et al., 2015). In the subspace $\bar{\mathcal{H}}$, spanned by N ($N > m + n$) linearly independent basis functions, of $\mathcal{H}$, the lifting function is given by $\psi = [\psi_1(x, u), \dots, \psi_N(x, u)]$. For convenience, the first m basis functions are defined as $\psi_i(x, u) = x_i$ (for $i = 1, 2, \dots, m$), where x_i denotes the i^{th} element of $x(t)$, and the last n basis functions are chosen as $\psi_i(x, u) = u_{i-N+n}$ (for $i = N - n + 1, \dots, N$). Accordingly, the lifting function evolves in the subspace as follows:

$$\dot{\psi} = \frac{d}{dt}\psi = \bar{\mathcal{K}}\psi \quad (15)$$

where $\bar{\mathcal{K}} \in \mathbb{R}^{N \times N}$ denotes the finite-dimensional approximation of $\mathcal{K}$. Then, we denote the first $N - m$ elements of ψ as z , implying $\psi = [z^T, u^T]^T$. Consequently, z is taken as the state of the nonlinear system in the lifted space. Thus, we can rewrite (15) as

$$\begin{bmatrix} \dot{z} \\ \dot{u} \end{bmatrix} = \bar{\mathcal{K}} \begin{bmatrix} z \\ u \end{bmatrix}. \quad (16)$$

To establish an explicit linear model, the approximated Koopman operator is partitioned as

$$\bar{\mathcal{K}} = \begin{bmatrix} \mathcal{K}_{zz} & \mathcal{K}_{zu} \\ \mathcal{K}_{uz} & \mathcal{K}_{uu} \end{bmatrix}. \quad (17)$$

Finally, we can learn an approximately equivalent linear model in the high-dimensional space for the nonlinear system (1) as follows:

$$\begin{aligned} \dot{z} &= Az + Bu, \\ x &= Cz \end{aligned} \quad (18)$$

where $A = \mathcal{K}_{zz}$, $B = \mathcal{K}_{zu}$, and $C = [I_{m \times m}, O_{m \times (N-m-n)}]$ with I being the identity matrix and O being a zero matrix.

Before exploiting the linear model (18) to design a controller, it is necessary to determine the remaining $N - m - n$ elements of the lifting function ψ and the system matrices A and B . Since selecting the lifting function manually to maintain prediction performance is a tough task (Shi and Meng, 2022), it would be preferable to learn the lifting function and system matrices simultaneously using a deep neural network, termed DKN. As illustrated in Figure 2, the DKN is composed of a lifting network and an evolution network, responsible for learning the lifting function and the system matrices, respectively. To enhance the ability to learn nonlinearity, rectified linear unit (ReLU) activation functions (formulated as $\text{ReLU}(s) = \max(0, s)$) are adopted in the output neurons of the lifting network. For convenience, we denote the output of the lifting network as $z_{\text{lift}}(x)$, the output of the DKN as $\hat{z}$, and the predicted state in the lifted space as $\hat{z}$. Considering the definitions of ψ and z , the original system state x is embedded in the first m elements of z . To train the DKN, the following K -step loss function is defined:

$$\begin{aligned} \mathcal{L} = \sum_{j=1}^K \beta^{j-1} & \left(\alpha_1 \mathcal{E}(\hat{z}_j, z_{\text{lift}}(\hat{x}_j)) + \alpha_2 \mathcal{E}(\hat{z}_j, z_{\text{lift}}(x_j)) \right. \\ & \left. + \alpha_3 \mathcal{E}(\hat{x}_j, x_j) + \alpha_4 \mathcal{E}(\dot{\hat{x}}_j, \dot{x}_j) \right) \end{aligned} \quad (19)$$

where hyperparameters $\alpha_1, \alpha_2, \alpha_3$, and α_4 are used to adjust the weights of different error items, β is the weight decay hyperparameter, $\mathcal{E}$ denotes the mean square error between two vectors, the subscript j indicates the values of associated variables at the j^{th} step, and $\hat{z}_j$ is calculated as

$$\hat{z}_j = \begin{cases} z_{\text{lift}}(x_j), & j = 1 \\ \hat{z}_{j-1} + \tau \dot{\hat{z}}_{j-1}, & j > 1 \end{cases} \quad (20)$$

with τ being the sampling interval of time when collecting data. To train the DKN, we collected data in the format of $\{(t_j, x_j, u_j, \dot{x}_j, \dot{u}_j) | j = 1, \dots, K\}$, where $t_j = j\tau$, by feeding the input sequence $\{u_j | j = 1, \dots, K\}$ to the nonlinear system formulated by (1) and measuring the system output. To enhance the prediction accuracy of the DKN, delays can be incorporated into the training data, as detailed in Bruder et al. (2019).

3.2. Koopman-operator-based iterative learning control

In this part, we elaborate the second contribution of this work. Since it is challenging to directly apply the AILC method in the original task space of continuum robots, we propose to perform iterative learning control in lifted space, which is achieved by improving the AILC controller (4) based on the learned continuous-time model (18).

Since (18) can only characterize the nonlinear system (1) approximately, we further complete (18) as follows by considering various uncertainties:

$$\dot{z}_k = Az_k + B((I + \Pi_k)u_k + d_k) + w_k \quad (21)$$

where $\Pi_k \in \mathbb{R}^{n \times n}$, $d_k \in \mathbb{R}^n$, and $w_k \in \mathbb{R}^{N-n}$ denote the input distribution uncertainty, the matched system uncertainty, and the unmatched system uncertainty in the k^{th} iteration, respectively, which are not learned by the Koopman network. Note that these uncertainties are state-dependent and time-varying. The current work aims at determining a sequence of control input $u_k(t)$, $\forall t \in [0, T_d]$, so that the lifted system state $z_k(t)$ can track the desired trajectory $z_d(t)$, obtained by lifting the reference state $x_d(t)$, gradually as the iteration number k increases. Accordingly, the tracking error in the lifted space is defined as $e_k(t) = z_k(t) - z_d(t)$. Notably, the task-space tracking error is included in $e_k(t)$ because the first m elements of $z_k(t)$ denote the robot state $x(t)$ according to the definition of the lifting function ψ . Next, according to Li and Huang (2024), we impose the following assumptions on (21).

Assumption 1. There exist some constants $\epsilon > 0$, $\eta > 0$, and $\zeta > 0$ such that $\|\Pi_k\| \leq \epsilon$, $\|d_k\| \leq \eta$, and $\|w_k\| \leq \zeta$, namely, the uncertainties are norm-bounded.

Assumption 2. For the input distribution uncertainty Π_k , there exists a constant $\xi \in (0, 1)$ satisfying $z^T(\xi I + \Pi_k)z \geq 0$, $\forall z \in \mathbb{R}^{N-n}$.

Assumption 3. It is assumed that the identical initialization condition, which is a common assumption in ILC area, is satisfied, namely, $z_k(0) = z_d(0)$, $\forall k \in \mathbb{N}$.

Remark 1. Assumption 1 implies that the uncertainties in system (21) are bounded, which is a common assumption in iterative learning control (Pierallini et al., 2021; 2023a). The results in the following section indicate that the model learned using the deep Koopman network can effectively predict the robot dynamics, which suggests that the learned model closely approximates the actual robot model, making the assumption of bounded model uncertainties reasonable. Assumption 2 implies that the control direction of the system (21) is determined. The control direction is implicitly contained in the input distribution matrix B , which can be learned from data. Therefore, this assumption is also reasonable. In

practice, when the starting point of the desired path is set to the end-effector position, Assumption 3 will hold. Therefore, when planning the path, setting the end effector as the starting point of the path can ensure that Assumption 3 holds. According to Li and Huang (2024), Assumption 3 can also be eliminated by incorporating initial state learning/correction techniques.

Then, the following error dynamics is derived:

$$\dot{e}_k = Ae_k + B((I + \Pi_k)u_k + d_k) + w_k - \dot{z}_d. \quad (22)$$

To achieve trajectory tracking tasks with perfect performance while compensating for uncertainties, we modify the AILC method (4) to the following form:

$$u_k = \begin{cases} u_{1,k} + u_{2,k} + u_{3,k} + u_{4,k}, & \|e_k^T B\| \neq 0 \\ u_{1,k}, & \|e_k^T B\| = 0 \end{cases} \quad (23)$$

where

$$u_{1,k} = \frac{B^T}{\|BB^T\|} (-Ke_k - Az_k + \dot{z}_d), \quad (24)$$

$$u_{2,k} = -\frac{\hat{\lambda}_k B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|Ke_k + Az_k - \dot{z}_d\| \|e_k\|, \quad (25)$$

$$u_{3,k} = -\frac{\hat{\eta}_k B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|e_k^T B\|, \quad (26)$$

$$u_{4,k} = -\frac{\hat{\zeta}_k B^T e_k}{(1 - \xi)\|e_k^T B\|^2} e_k^T \text{sign}(e_k), \quad (27)$$

with $K \in \mathbb{R}^{(N-n) \times (N-n)}$ being a positive-definite feedback gain matrix. Besides, $\hat{\lambda}_k$ is responsible for learning the value of $\lambda = \|B\|^2 (1 + \epsilon)/\|BB^T\| + 1$, while $\hat{\eta}_k$ and $\hat{\zeta}_k$ are responsible for learning the values of η and ζ , respectively. Notably, $u_{2,k}$, $u_{3,k}$, and $u_{4,k}$ determine the robustness of the proposed control scheme regarding external disturbances and system uncertainties. $u_{2,k}$ handles the input distribution uncertainty, $u_{3,k}$ compensates for the matched system uncertainty d_k , and $u_{4,k}$ addresses the mismatched disturbance w_k . Accordingly, the parameter updating laws (10)–(12) are also modified as follows:

$$\hat{\lambda}_k = \begin{cases} 0, & k = -1 \\ \hat{\lambda}_{k-1} + \gamma_1 \|Ke_k + Az_k - \dot{z}_d\| \|e_k\|, & k \geq 0 \end{cases}, \quad (28)$$

$$\hat{\eta}_k = \begin{cases} 0, & k = -1 \\ \hat{\eta}_{k-1} + \gamma_2 \|e_k^T B\|, & k \geq 0 \end{cases}, \quad (29)$$

$$\hat{\zeta}_k = \begin{cases} 0, & k = -1 \\ \hat{\zeta}_{k-1} + \gamma_3 e_k^T \text{sign}(e_k), & k \geq 0 \end{cases}, \quad (30)$$

where γ_1 , γ_2 , and γ_3 are learning rates to be designed. It is worth noting that numerical singularities may occur when $\|e_k^T B\|$ is very close to zero. To avoid the problem, one can

replace $\|e_k^T B\|$ with ρ when $\|e_k^T B\| < \rho$, where $\rho > 0$ is a sufficiently small constant.

The differences between the AILC method and our proposed KILC method can be summarized as follows. First, the function $f(x_k)$ and input distribution matrix B in the AILC method need to be derived from the analytical model of the system, while the system matrix A and input matrix B in our KILC method are learned from data via the DKN. Second, the unknown parametric uncertainty Θ in the AILC method does not need to be considered in our KILC method because the system matrix A and system state z_k are known, thereby eliminating the parameter learning law (9). Third, the first-order time derivative of the reference system state is not included in the error dynamics (3) associated with the AILC method, while $\dot{z}_d$ is introduced in the error dynamics (22). All these differences result in the modifications in terms of control laws and parameter learning laws.

3.3. Theoretical analysis

Finally, we analyze the convergence and robustness of our KILC method, which is the third contribution of this work. The analysis is inspired by Li and Huang (2024) but the details are different, primarily due to the differences in the error dynamics and learning laws.

Theorem 1. *When the assumptions 1, 2, and 3 are satisfied by the dynamic system (21), the proposed KILC algorithm composed of the controller (23) and the parameter learning laws (28)–(30) guarantees that the tracking error e_k converges as the iteration number k increases, that is, $\lim_{k \rightarrow \infty} e_k(t) = 0$, $\forall t \in [0, T_d]$.*

Proof. To begin with, a composite energy function is defined as

$$E_k(t) = \frac{e_k^T e_k}{2} + \frac{\int_0^t \tilde{\lambda}_k^2 d\tau}{2\gamma_1} + \frac{\int_0^t \tilde{\eta}_k^2 d\tau}{2\gamma_2} + \frac{\int_0^t \tilde{\zeta}_k^2 d\tau}{2\gamma_3} \quad (31)$$

where $\tilde{\lambda}_k = \hat{\lambda}_k - \lambda$, $\tilde{\eta}_k = \hat{\eta}_k - \eta$, and $\tilde{\zeta}_k = \hat{\zeta}_k - \zeta$ are parameter estimation errors. The proof involves the following two steps.

Step 1: The difference of E_k . The difference of E_k along the iteration axis is defined as

$$\begin{aligned} \Delta E_k &= E_k - E_{k-1} \\ &= \frac{e_k^T e_k}{2} - \frac{e_{k-1}^T e_{k-1}}{2} + \frac{\int_0^t \tilde{\lambda}_k^2 - \tilde{\lambda}_{k-1}^2 d\tau}{2\gamma_1} \\ &\quad + \frac{\int_0^t \tilde{\eta}_k^2 - \tilde{\eta}_{k-1}^2 d\tau}{2\gamma_2} + \frac{\int_0^t \tilde{\zeta}_k^2 - \tilde{\zeta}_{k-1}^2 d\tau}{2\gamma_3}. \end{aligned} \quad (32)$$

By substituting the controller (23) into the error dynamics (22), we have

$$\begin{aligned} \dot{e}_k &= Az_k - \dot{z}_d + \frac{B(I + \Pi_k)B^T}{\|BB^T\|} q_k \\ &\quad - \frac{\hat{\lambda}_k B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|q_k\| \|e_k\| \\ &\quad - \frac{\hat{\eta}_k B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|e_k^T B\| \\ &\quad - \frac{\hat{\zeta}_k B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} e_k^T \text{sign}(e_k) \\ &\quad + Bd_k + w_k \end{aligned} \quad (33)$$

where $q_k = -Ke_k - Az_k + \dot{z}_d$. Subsequently, the following result is obtained:

$$\begin{aligned} e_k^T \dot{e}_k &= -e_k^T Ke_k + e_k^T \left(\frac{B(I + \Pi_k)B^T}{\|BB^T\|} - I \right) q_k \\ &\quad - \frac{\hat{\lambda}_k e_k^T B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|q_k\| \|e_k\| \\ &\quad - \frac{\hat{\eta}_k e_k^T B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \|e_k^T B\| \\ &\quad - \frac{\hat{\zeta}_k e_k^T B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} e_k^T \text{sign}(e_k) \\ &\quad + e_k^T Bd_k + e_k^T w_k. \end{aligned} \quad (34)$$

As a result of Assumption 2 and the equality $I + \Pi_k = I - \xi I + \xi I + \Pi_k$, the following inequality is obtained:

$$-\frac{e_k^T B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \leq -\|e_k^T B\|, \quad (35)$$

which leads to

$$-\frac{e_k^T B(I + \Pi_k)B^T e_k}{(1 - \xi)\|e_k^T B\|^2} \leq -1. \quad (36)$$

Besides, we have

$$\begin{aligned} &e_k^T \left(\frac{B(I + \Pi_k)B^T}{\|BB^T\|} - I \right) q_k \\ &\leq \|e_k\| \left(\frac{\|B\|^2(1 + \epsilon)}{\|BB^T\|} + 1 \right) \|q_k\| = \lambda \|q_k\| \|e_k\|. \end{aligned} \quad (37)$$

Substituting (36) and (37) into (34) yields

$$\begin{aligned} e_k^T \dot{e}_k &\leq -e_k^T Ke_k - \tilde{\lambda}_k \|q_k\| \|e_k\| - \hat{\eta}_k \|e_k^T B\| \\ &\quad - \hat{\zeta}_k e_k^T \text{sign}(e_k) + e_k^T Bd_k + e_k^T w_k \\ &\leq -e_k^T Ke_k - \tilde{\lambda}_k \|q_k\| \|e_k\| - \hat{\eta}_k \|e_k^T B\| \\ &\quad - \hat{\zeta}_k e_k^T \text{sign}(e_k) + \eta \|e_k^T B\| + \zeta e_k^T \text{sign}(e_k) \\ &= -e_k^T Ke_k - \tilde{\lambda}_k \|q_k\| \|e_k\| - \tilde{\eta}_k \|e_k^T B\| \\ &\quad - \tilde{\zeta}_k e_k^T \text{sign}(e_k). \end{aligned} \quad (38)$$

Accordingly, we have

$$\frac{e_k^T e_k}{2} = \int_0^t e_k^T(\tau) \dot{e}_k(\tau) d\tau \leq - \int_0^t e_k^T K e_k + \quad (39)$$

$$\tilde{\lambda}_k \|q_k\| \|e_k\| + \tilde{\eta}_k \|e_k^T B\| + \tilde{\zeta}_k e_k^T \text{sign}(e_k) d\tau,$$

where Assumption 3 is applied. For the third item of (32), we have

$$\begin{aligned} & \frac{1}{2\gamma_1} \int_0^t \tilde{\lambda}_k^2 - \tilde{\lambda}_{k-1}^2 d\tau \\ &= \frac{1}{2\gamma_1} \int_0^t \left(-2\tilde{\lambda}_k + \hat{\lambda}_k - \hat{\lambda}_{k-1} \right) \left(\hat{\lambda}_{k-1} - \hat{\lambda}_k \right) d\tau \\ &= \int_0^t \tilde{\lambda}_k \|q_k\| \|e_k\| d\tau - \frac{1}{2\gamma_1} \int_0^t \left(\hat{\lambda}_k - \hat{\lambda}_{k-1} \right)^2 d\tau \\ &\leq \int_0^t \tilde{\lambda}_k \|q_k\| \|e_k\| d\tau. \end{aligned} \quad (40)$$

Likewise, we also have

$$\frac{1}{2\gamma_2} \int_0^t \tilde{\eta}_k^2 - \tilde{\eta}_{k-1}^2 d\tau \leq \int_0^t \tilde{\eta}_k \|e_k^T B\| d\tau \quad (41)$$

and

$$\frac{1}{2\gamma_3} \int_0^t \tilde{\zeta}_k^2 - \tilde{\zeta}_{k-1}^2 d\tau \leq \int_0^t \tilde{\zeta}_k e_k^T \text{sign}(e_k) d\tau. \quad (42)$$

By combining (32), (39), (40), (41), and (42), we can finally get

$$\Delta E_k \leq - \int_0^t e_k^T K e_k d\tau - \frac{e_{k-1}^T e_{k-1}}{2} \leq - \frac{\|e_{k-1}\|^2}{2}. \quad (43)$$

Step 2: The convergence of the tracking error. Based on (43), we have

$$E_k(t) \leq E_0(t) - \frac{\sum_{i=1}^k \|e_{i-1}\|^2}{2}. \quad (44)$$

Since $E_k(t)$ is positive, one can determine that $\lim_{k \rightarrow \infty} e_k(t) = 0$ if $E_0(t)$ is bounded. Finally, we need to prove the boundedness of $E_0(t)$. According to (38), we have

$$\begin{aligned} \dot{E}_0(t) &= e_0^T \dot{e}_0 + \frac{\tilde{\lambda}_0^2}{2\gamma_1} + \frac{\tilde{\eta}_0^2}{2\gamma_2} + \frac{\tilde{\zeta}_0^2}{2\gamma_3} \\ &\leq -e_0^T K e_0 - \tilde{\lambda}_0 \|q_0\| \|e_0\| - \tilde{\eta}_0 \|e_0^T B\| \\ &\quad - \tilde{\zeta}_0 e_0^T \text{sign}(e_0) + \frac{\tilde{\lambda}_0^2}{2\gamma_1} + \frac{\tilde{\eta}_0^2}{2\gamma_2} + \frac{\tilde{\zeta}_0^2}{2\gamma_3} \\ &= -e_0^T K e_0 - \frac{\tilde{\lambda}_0 \hat{\lambda}_0}{\gamma_1} - \frac{\tilde{\eta}_0 \hat{\eta}_0}{\gamma_2} - \frac{\tilde{\zeta}_0 \hat{\zeta}_0}{\gamma_3} \\ &\quad + \frac{\tilde{\lambda}_0^2}{2\gamma_1} + \frac{\tilde{\eta}_0^2}{2\gamma_2} + \frac{\tilde{\zeta}_0^2}{2\gamma_3} \\ &= -e_0^T K e_0 - \frac{\tilde{\lambda}_0^2}{2\gamma_1} - \frac{\tilde{\lambda}_0}{\gamma_1} - \frac{\tilde{\eta}_0^2}{2\gamma_2} - \frac{\tilde{\eta}_0}{\gamma_2} \\ &\quad - \frac{\tilde{\zeta}_0^2}{2\gamma_3} - \frac{\tilde{\zeta}_0}{\gamma_3}. \end{aligned} \quad (45)$$

By applying the Young's inequality, we have

$$-\frac{\tilde{\lambda}_0}{\gamma_1} \leq \frac{\tilde{\lambda}_0^2}{2\gamma_1} + \frac{\lambda^2}{2\gamma_1}, \quad -\frac{\tilde{\eta}_0}{\gamma_2} \leq \frac{\tilde{\eta}_0^2}{2\gamma_2} + \frac{\eta^2}{2\gamma_2}, \quad -\frac{\tilde{\zeta}_0}{\gamma_3} \leq \frac{\tilde{\zeta}_0^2}{2\gamma_3} + \frac{\zeta^2}{2\gamma_3}. \quad (46)$$

Then, we can further get

$$\dot{E}_0(t) \leq -e_0^T K e_0 + \frac{\lambda^2}{2\gamma_1} + \frac{\eta^2}{2\gamma_2} + \frac{\zeta^2}{2\gamma_3}. \quad (47)$$

Since $\lambda, \eta, \zeta, \gamma_1, \gamma_2,$ and γ_3 are constants, it is possible to find a sufficiently large gain matrix K to ensure $e_0^T K e_0 > \lambda^2/2\gamma_1 + \eta^2/2\gamma_2 + \zeta^2/2\gamma_3$, which means $\dot{E}_0(t)$ is negative, implying $E_0(t)$ is bounded over $[0, T_d]$. The proof is thus completed.

4. Results

The performance of the proposed control method was evaluated using both a motor-driven continuum robot and a pneumatic continuum robot, as depicted in Figure 2. To highlight the advantages of the proposed KILC method, we compared it against the KMPC method (Bruder et al., 2019) and the KLQR method (Haggerty et al., 2023). The primary results are detailed as follows.

4.1. Setup

The experimental setup includes two continuum robots, a visual sensor, and a personal computer (with Intel(R) Core(TM) i5-9400 CPU@2.90GHz). The Micron Tracker H3-60 sensor measures the tip position using a marker attached to the robot tip. The control algorithms are implemented in Simulink.

The motor-driven continuum robot has a length of $L = 260$ mm, actuated by five motors. The base motor is responsible for translating the robot. The other four motors are divided into two pairs, each responsible for bending the robot within a plane. Due to the actuation coupling between opposite bending motors, the robot has three DoF in the actuation space. To train the DKN, about 2600 pairs of data were collected by controlling the robot to explore its workspace. For the DKN, the size of the hidden layer in the lifting network was 128 and the size of the lifted space was set to 30. The hyperparameters of the loss function were set as $\alpha = 0.5, \alpha_2 = \alpha_3 = \alpha_4 = 1, \beta = 0.9$, and $K = 10$. For the proposed KILC controller, the feedback gain matrix was set to $K = 10I$, the parameter ξ was set to 0.1, and the learning rates were set to $\gamma_1 = \gamma_2 = \gamma_3 = 0.01$. One time delay was incorporated to train the Koopman operator with better accuracy. For the KMPC method (Bruder et al., 2019), a set of lifting functions ($N = 55$), consisting of all polynomials of maximum degree 2, were used when identifying the Koopman model with one time delay. The prediction horizon of the MPC controller was set to 10 time steps, with the running and terminal costs being 0.1 and 100, respectively. For the KLQR method (Haggerty et al., 2023), it was reported that time delay-only lifting functions offer better results than high-order functions. Therefore, four time

delays were included in the lifting functions when identifying the Koopman model, leading to $N = 16$ lifting functions. When calculating the gain matrix for the LQR controller, the input cost matrix and state cost matrix were set to the identity matrix I . In addition, these two methods were also trained on 2600 pairs of data.

The pneumatic continuum robot has a length of $L = 500$ mm. It is composed of two segments, each of which is actuated by three pneumatic muscles. To train the DKN, about 25,600 pairs of data were collected by controlling the robot to explore its workspace. Since its actuation space is more complex than the motor-driven continuum robot, it is necessary to collect more data. For the DKN, the size of the hidden layer in the lifting network was 128 and the size of the lifted space was set to 50. For the proposed KILC controller, the feedback gain matrix was set to $K = 5I$, the parameter ξ was set to 0.1, and the learning rates were set to $\gamma_1 = \gamma_2 = \gamma_3 = 0.0001$. Due to the instability of the robot, the feedback gain and learning rates were set smaller. The parameters of the KMPC method (Bruder et al., 2019) and the KLQR method (Haggerty et al., 2023) were the same as those for the motor-driven continuum robot.

4.2. Experiments on the motor-driven continuum robot

First, we evaluate the performance of the proposed DKN in learning the robot model. The DKN receives the robot state (e.g., the tip position of the robot) and actuation signal as input and outputs the predicted tip velocity. It was trained on 2600 data pairs, and the training process took approximately 13 minutes, covering 4800 iterations (60 epochs). As shown in Figure 3(a), the training loss decreased rapidly during the first 1000 iterations and then gradually declined in the subsequent iterations. Subsequently, the well-trained DKN was tested on a separate dataset to evaluate its prediction ability. Figure 3(b) presents a comparison between the actual tip velocity of the robot and the predicted velocity along different axes in the Cartesian space. Despite fluctuations in the collected data due to the robot's imperfections, the DKN demonstrated satisfactory prediction accuracy, highlighting its effectiveness. Besides, we also compared the one-step prediction performance among linear models learned by KMPC, KLQR, and KILC methods. As shown in Figure 4, there are no major differences from the point of view of the modeling. The modeling error of our KILC method is even slightly larger than the KLQR method, but the control performance of our KILC method is much better as elaborated below.

Subsequently, using the learned model, the robot was controlled via our KILC method to track a Pac-Man trajectory across various scenarios. Five scenarios, including ideal environment, motor fault, input disturbance, external disturbance, and less-explored area, were designed to evaluate the performance of the proposed KILC method and compare it against other control methods in terms of accuracy, efficiency, robustness, and generalizability. The

robot was controlled using different methods to track the same reference trajectory in these scenarios.

In the ideal environment, the reference trajectory falls entirely within a fully explored area during training, and the robot is not subject to disturbances. In this case, the detailed experiment results are presented in Figure 5, including snapshots of the robot (Figure 5(a)) and the comparison between the reference trajectory and the actual robot tip trajectory generated by our KILC method in the first iteration in the Cartesian space (Figure 5(b)) and along different axes (Figure 5(c)), which confirm the effectiveness of our method. Figure 5(d) illustrates a box plot of tracking error over iterations for our KILC method, indicating a reduction in tracking error through iterative learning. As depicted in Figure 5(e), to compare the computational efficiency of different methods, the average time required to obtain the solution per time step was recorded. The computational time of our KILC method (0.32 ms in the first iteration) is comparable to the KLQR method (0.27 ms), and both methods require significantly less time than the KMPC method (4.65 ms), highlighting the computational efficiency of our approach. The tracking errors across different scenarios for each method are summarized in Figure 6, and the specific trajectory comparisons are summarized in Figure 7. It is evident from Figure 6(b) that, in the ideal environment, the KLQR method exhibited the largest tracking error, while our KILC method achieved slightly lower tracking error in the first iteration compared to the KMPC method, underscoring the accuracy of our proposed method. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S1](#).

Then, we assessed the fault tolerance capability of different methods by disabling the base motor of the robot, resulting in the loss of one DoF in the actuation space of the robot. Due to the actuation coupling of the motors, the robot only has three DoF in total and the loss of one actuation DoF will inevitably lead to the failure of the tracking task in the three-dimensional (3D) Cartesian space, regardless of control algorithm improvements. Therefore, to assess the ability of the control methods in dealing with motor fault utilizing actuation redundancy, the robot was only required to complete the tracking task in a two-dimensional (2D) plane in this scenario. Without motor fault, the KMPC, KLQR, and KILC (in the first iteration) methods achieved the root mean square errors (RMSEs) of 1.85 mm, 4.94 mm, and 0.89 mm, respectively. In the presence of motor fault, these methods achieved the RMSEs of 2.34 mm, 5.00 mm, and 0.91 mm, respectively. It is evident that the negative impact of the motor fault on our KILC method is minimal. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S2](#).

Next, to assess the input disturbance rejection capabilities of different methods, a sinusoidal disturbance was introduced to the bending motors. The resulting tracking errors are illustrated in Figure 6(b). Compared to the ideal

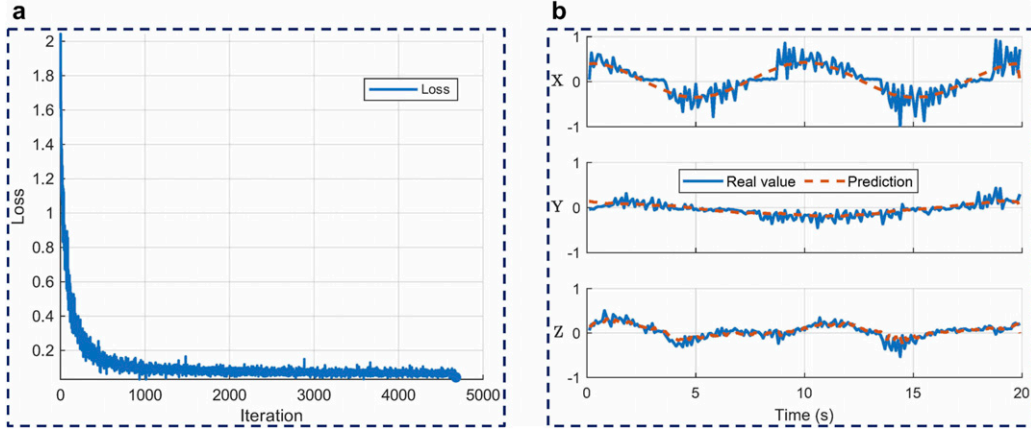


Figure 3. Performance evaluation of the DKN. (a) Profile of the loss function over the iteration number during the training of the DKN using data collected from the motor-driven continuum robot. (b) Comparison between the normalized velocities of the robot tip predicted by the DKN and the actual normalized velocities along X-, Y-, and Z-axes in Cartesian space.

environment, the KMPC and KLQR methods exhibited significant increases in error due to the disturbance, while the negative impact on our KILC method (in both the first and fifth iterations) remained minimal. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S3](#). In the presence of external disturbance (i.e., an obstacle that did not exist in the previous training stage), similar results were observed. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S4](#).

When collecting training data, it is challenging to fully traverse the robot's entire workspace. To validate the generalizability of the three aforementioned control methods, the robot was controlled to track a trajectory that passes through the less-explored area, as illustrated in [Figure 6\(b\)](#). It was observed that the tracking errors of the KMPC and KLQR methods showed significant increases, whereas the tracking errors of our KILC method (in both the first and fifth iterations) were almost not affected. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S5](#).

Finally, the numerical results of the various methods in the aforementioned experiments are summarized in [Table 1](#). For a better comparison, as illustrated in [Figure 6\(c\)](#), a radar graph visualizes the performance of the three aforementioned control methods. Six metrics, calculated using the formulas shown in [Table 2](#), were adopted to evaluate these methods, corresponding to the above five scenarios and the time cost. The accuracy is determined by the root mean square error (RMSE) in the tracking control task. The computational efficiency is evaluated based on the average time required to calculate the control signal per time step. Other metrics are determined by the increase in RMSE under various scenarios relative to the ideal environment. The metric values increase from 0 to 1 as they move from the center to the periphery of the graph, where the periphery

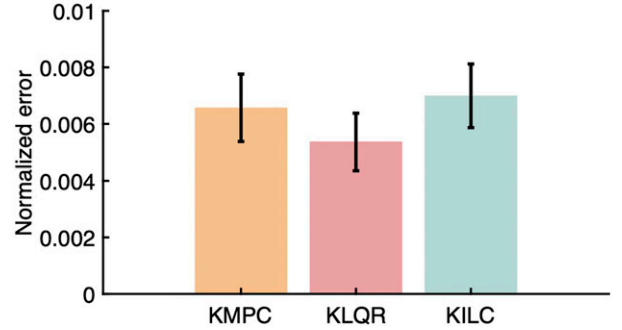


Figure 4. Comparison of one-step prediction performance among linear models learned by different methods. Since the learned linear model in our KILC method predicts velocity information instead of position information, we adopted the one-step Euler method to further predict position information, which is consistent with the other two methods.

indicates better performance. Compared to the KMPC method, the KLQR method performs worse in terms of accuracy but significantly better in other metrics, while our KILC method (in both the first and fifth iterations) outperforms the KMPC method across all metrics. Compared to the KLQR method, our KILC method (in both the first and fifth iterations) shows similar performance in computational efficiency and fault tolerance but achieves better results in other aspects. Our KILC method achieves satisfactory performance across all six metrics in the very first iteration, with further improvements after subsequent iterations. In summary, these experimental results confirm the accuracy, efficiency, and significantly enhanced robustness and generalizability of our proposed KILC method.

4.3. Experiments on the pneumatic continuum robot

Then, physical experiments were conducted on a pneumatic continuum robot. Similarly, the robot was controlled using

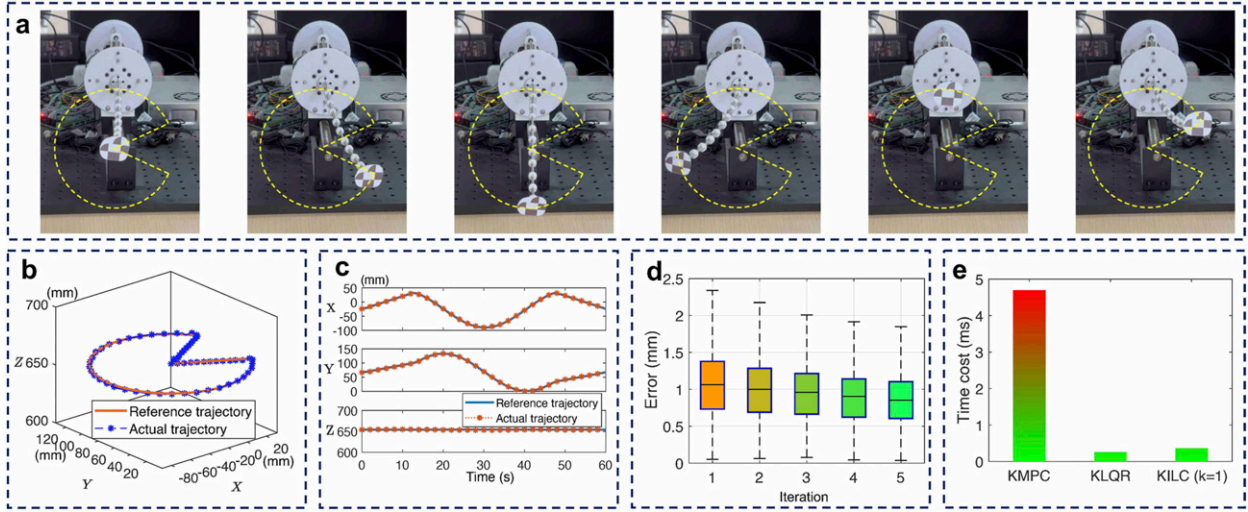


Figure 5. Tracking control results of the motor-driven continuum robot in ideal environment. (a) Snapshots of the robot tracking a Pac-Man trajectory achieved by our KILC method in the first iteration. (b) Comparison between the reference trajectory x_d and the robot tip trajectory x generated by our KILC method in the first iteration. (c) Comparison between the reference trajectory and the robot tip trajectory with respect to X -, Y -, and Z -axes. (d) Box plot of tracking error with respect to the iteration number of our KILC method. (e) Average time required by different methods to calculate the control signal per time step.

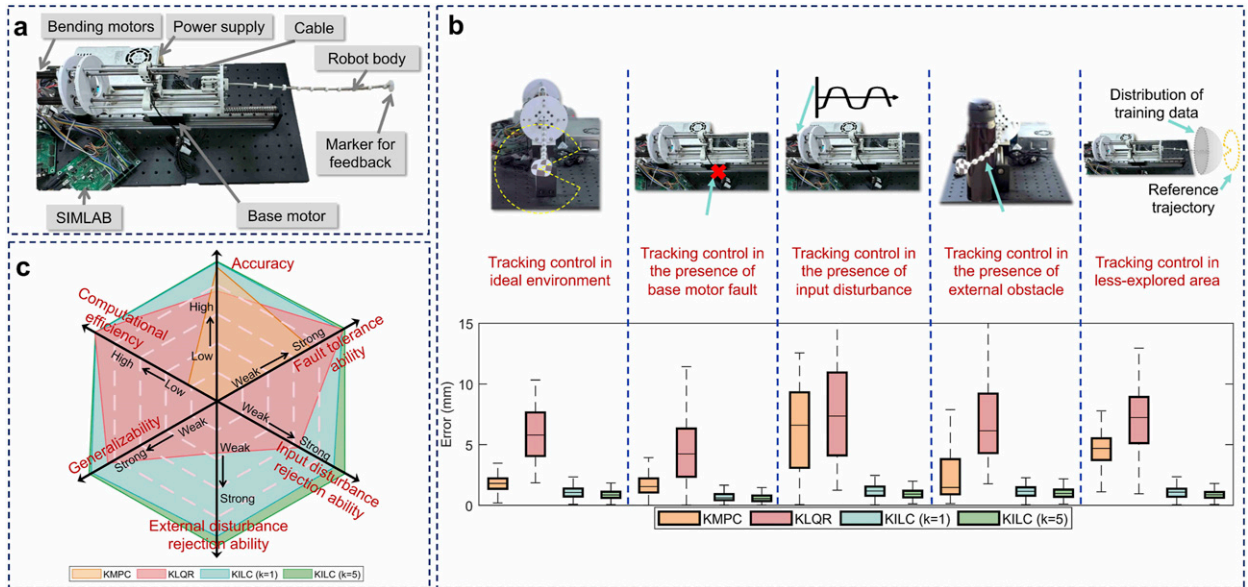


Figure 6. Comparison with the KMPC and the KLQR methods for controlling the motor-driven continuum robot. (a) Components of the physical robot platform. (b) Box plots of tracking errors associated with different methods in various scenarios, including ideal environment, motor fault, input disturbance, external disturbance, and less-explored area. The symbol k indicates the iteration number of our KILC method. (c) Performance overview of different methods across six metrics.

various methods to track a Pac-Man trajectory in five different scenarios within the 3D Cartesian space.

In ideal environment, the reference trajectory falls entirely within a fully explored area during training, and the robot is not subject to disturbances. In this case, the detailed experiment results are presented in Figure 8, including snapshots of the robot (Figure 8(a)) and the comparison between the reference trajectory and the actual robot tip trajectory

generated by our KILC method in the third iteration in the Cartesian space (Figure 8(b)) and along different axes (Figure 8(c)), which confirm the effectiveness of our method. Figure 8(d) illustrates a box plot of tracking error over iterations for our KILC method, indicating a reduction in tracking error through iterative learning. As depicted in Figure 8(e), to compare the computational efficiency of different methods, the average time required to obtain the

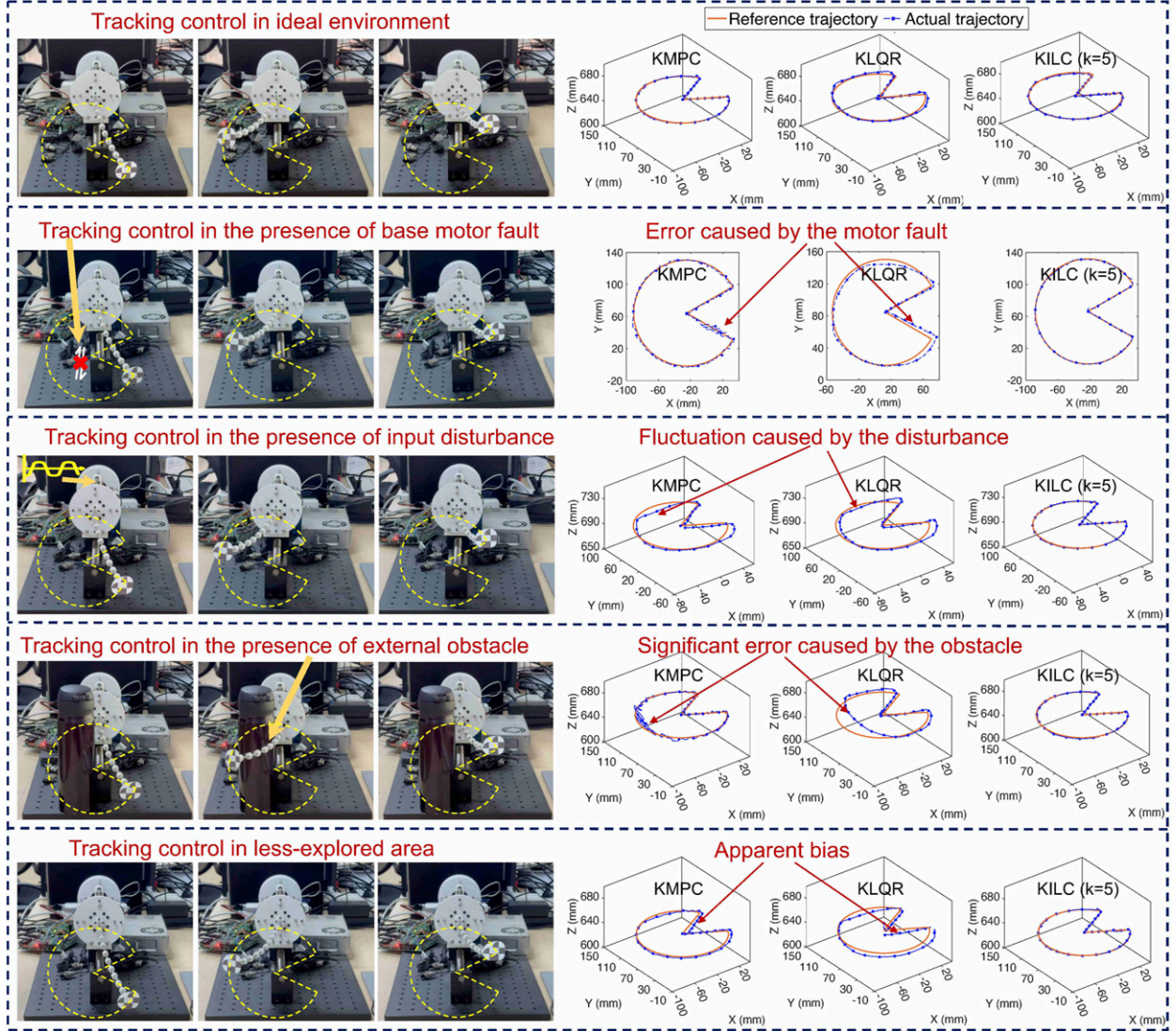


Figure 7. Specific experiment results on the motor-driven continuum robot, including snapshots of the robot (controlled by our KILC method in the fifth iteration) and trajectory comparisons for different methods.

Table 1. Tracking Errors and Time Costs Associated With Different Methods in the Tracking Task Performed by the Motor-Driven Continuum Robot. The Best Result in Each Column is in Bold.

Method	RMSE (mm)						Time cost (ms)
	Ideal environment	No motor fault (2D)	Motor fault (2D)	Input disturbance	External disturbance	Less-explored area	
KMPC	2.02	1.85	2.34	7.21	8.18	4.76	4.65
KLQR	6.28	4.94	5.00	8.54	10.32	7.20	0.27
KILC ($k = 1$)	1.16	0.89	0.91	1.26	1.24	1.18	0.32
KILC ($k = 5$)	0.97	0.79	0.77	1.14	1.12	0.96	0.30

solution per time step was recorded. The computational time of our KILC method (0.36 ms in the first iteration) is comparable to the KLQR method (0.30 ms), and both methods require significantly less time than the KMPC

method (8.1 ms), highlighting the computational efficiency of our approach. The tracking errors across different scenarios for each method are summarized in Figure 9, and the specific trajectory comparisons are summarized in Figure 10.

As illustrated in Figure 9(b), the KLQR method resulted in a larger tracking error compared to the KMPC and KILC methods in the ideal environment. The RMSEs of these three methods were 8.90 mm, 6.45 mm, and 6.76 mm (in the first iteration of our KILC method), respectively. Although the KILC method initially exhibited a larger RMSE in the first iteration compared to the KMPC method, the RMSE was reduced to 6.11 mm after subsequent iterations ($k = 3$). The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S6](#).

Table 2. Performance Metrics and Their Associated Calculation Formulas for the Motor-Driven Continuum Robot. RMSE₁: RMSE in Ideal Environment. RMSE₂: RMSE in the Presence of Motor Fault. RMSE₃: RMSE in the Presence of Input Disturbance. RMSE₄: RMSE in the Presence of External Disturbance. RMSE₅: RMSE in Less-Explored Area. After the Metric of Accuracy, When Calculating the Metrics Associated With the Fifth Iteration of the KILC Method, We Use the Value of the RMSE₁ Associated With the First Iteration to Better Show the Improvement of Iterative Learning. The Numerical Results are Limited in [0,1].

Metrics	Formula
Accuracy	$1 - \min\left(1, \frac{\text{RMSE}_1}{5\%L}\right)$
Fault tolerance	$1 - \min\left(1, \frac{\text{RMSE}_2 - \text{RMSE}_1}{\text{RMSE}_1}\right)$
Input disturbance rejection	$1 - \min\left(1, \frac{\text{RMSE}_3 - \text{RMSE}_1}{\text{RMSE}_1}\right)$
External disturbance rejection	$1 - \min\left(1, \frac{\text{RMSE}_4 - \text{RMSE}_1}{\text{RMSE}_1}\right)$
Generalizability	$1 - \min\left(1, \frac{\text{RMSE}_5 - \text{RMSE}_1}{\text{RMSE}_1}\right)$
Computational efficiency	$1 - \frac{\text{Time cost}}{6}$

It is worth noting that, compared with the motor-driven continuum robot, the overall tracking error of the pneumatic continuum robot was larger, primarily due to the inherent limitations of the robot's precision. To evaluate the stability and precision of the robot prototype, as shown in Figure 11, smoothly varying actuation signals were applied to the robot, revealing that its end-effector jittered during the movement process, with a maximum jitter range of 16.5 mm and a mean jitter range of 3.4 mm. Given this limitation of the robot, the tracking errors achieved by these methods are considered acceptable.

Then, we verified the fault tolerance capability of different methods by disabling one of the pneumatic muscles of the robot as illustrated in Figure 9(b), resulting in the loss of one DoF in the actuation space of the robot. Since the two-segment continuum robot is actuated by six pneumatic muscles, it retains sufficient actuation DoF to complete the 3D tracking task despite the muscle fault. In this scenario, the KMPC, KLQR, and KILC (in the first iteration) methods achieved RMSEs of 8.55 mm, 13.43 mm, and 6.78 mm, respectively. Evidently, the muscle fault led to significant increases in the tracking errors of the KMPC and KLQR methods, while having a negligible effect on the KILC method in both the first and third iterations, indicating that the KILC method possesses a much higher fault tolerance capability than the compared methods. The trajectory tracking processes corresponding to different methods in this case are demonstrated in the [Supplemental Material Movie S7](#).

Next, to evaluate the input disturbance rejection ability of different methods, a sinusoidal input disturbance was added to one of the pneumatic muscles. Similarly, an obstacle was introduced into the environment to assess the

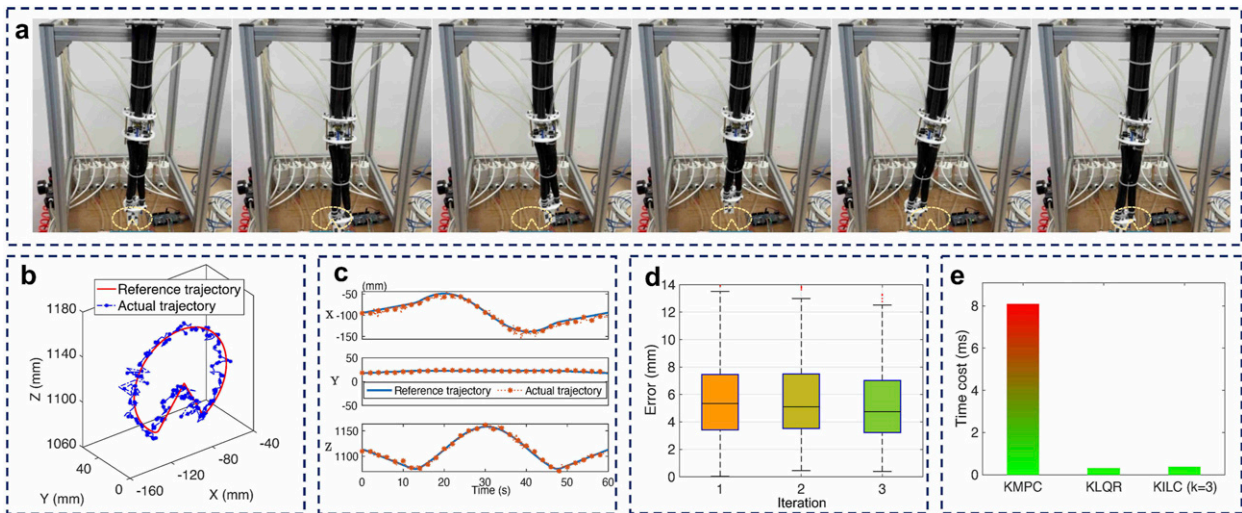


Figure 8. Tracking control results of the pneumatic continuum robot in ideal environment. (a) Snapshots of the robot tracking a Pac-Man trajectory achieved by our KILC method in the third iteration. (b) Comparison between the reference trajectory x_d and the robot tip trajectory x generated by our KILC method in the third iteration. (c) Comparison between the reference trajectory and the robot tip trajectory with respect to X-, Y-, and Z-axes. (d) Box plot of tracking error with respect to the iteration number of our KILC method. (e) Average time required by different methods to calculate the control signal per time step.

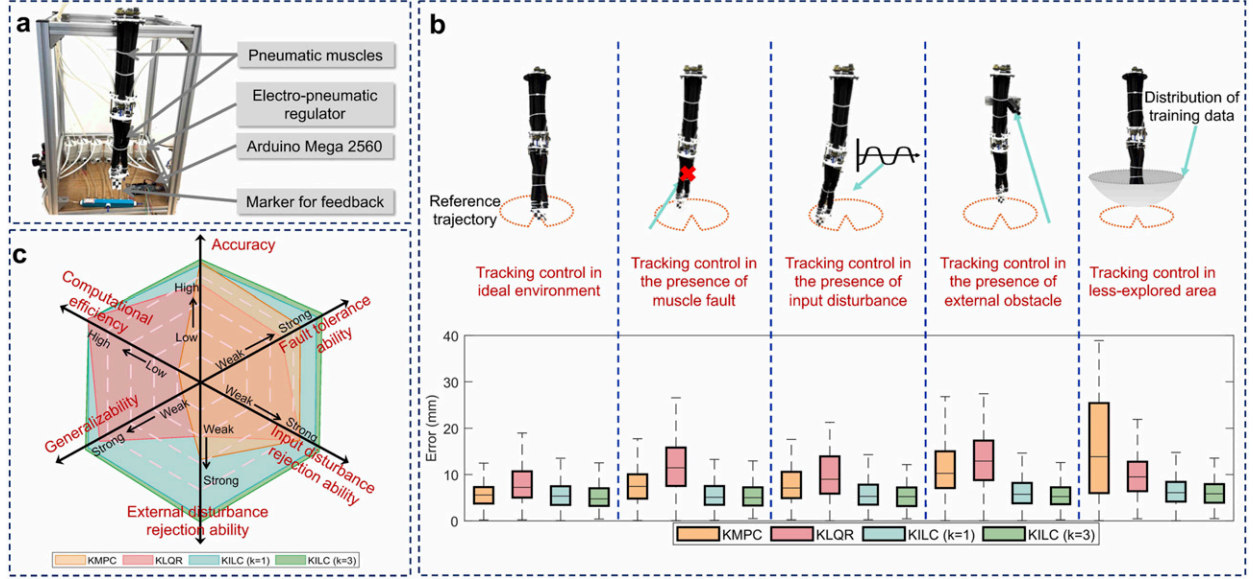


Figure 9. Comparison with the KMPC and the KLQR methods for controlling the pneumatic continuum robot. (a) Components of the physical robot platform. (b) Box plots of tracking errors associated with different methods in various scenarios, including ideal environment, motor fault, input disturbance, external disturbance, and less-explored area. (c) Performance overview of different methods in terms of six metrics.

external disturbance rejection capability of the methods. In addition, to validate the generalizability of these methods, the robot was controlled to track a trajectory that passes through the less-explored area. The box plots illustrating the tracking errors across these three scenarios are presented in Figure 9(b). In comparison to the ideal environment, the KMPC and KLQR methods exhibited significant increases in tracking errors across these scenarios, while our KILC method remained largely unaffected in both the first and third iterations. The trajectory tracking processes corresponding to different methods in these cases are demonstrated in Supplemental Material Movie S8, S9, and S10.

Finally, the numerical results of different methods from the aforementioned experiments are summarized in Table 3. To facilitate comparison, as illustrated in Figure 9(c), the radar graph visualizes the performance of the various methods. In summary, the KMPC method achieved performance comparable to our method in terms of accuracy, while the KLQR method demonstrated similar performance to our method in terms of computational efficiency. However, our KILC method surpassed both methods in all the other metrics in the first and third iterations. Our method achieved satisfactory performance in all the six metrics from the very first iteration, with further improvement observed after three iterations. These experimental results with the pneumatic continuum robot clearly confirmed the accuracy, efficiency, and significantly superior robustness and generalizability of our proposed KILC method.

Our validation results indicated that the modeling ability of our KILC method is slightly worse than the KMPC and KLQR methods in terms of accuracy, but the control performance of our KILC method is much better. In other words, the differences among these methods mainly stem

from the controllers. The superior performance of our KILC method is partially due to the use of high feedback gains. Additionally, the poor performance of KMPC and KLQR under non-ideal environments is not due to wrongly tuned parameters but rather inherent limitations of them. In the ideal environment, the models learned by the KMPC and KILC methods match the actual situation. Compared to non-ideal conditions, the tracking errors obtained by KMPC and KILC (in the first iteration) for both types of continuum robots are relatively close in the ideal environment. However, under non-ideal conditions, the learned models do not match well. In our KILC method, we overcome this issue by adopting high feedback gains, whereas the KMPC method fails to do so. The KMPC method lifts the feedback state to a higher-dimensional state as the initial high-dimensional state and predicts future states through the linear model to minimize the objective function. However, under non-ideal conditions, the predicted state of this model deviates significantly from the actual state, leading to control signals that fail to meet the desired requirements. From the KMPC formulation (Bruder et al., 2019), it is evident that only the positive definite matrices are adjustable parameters, and both are related solely to the predicted state or the input state, but not to feedback. Furthermore, the KMPC formulation does not explicitly consider feedback error, such as $e_k = z_k - z_d$ in our KILC method. This results in the absence of an adjustable feedback gain coefficient or parameter in the KMPC method. While it is promising to enhance KMPC by explicitly incorporating feedback terms, it is important to recognize that the core principle of MPC is to optimize control actions based on the predicted error between future robot states and desired trajectories. Nevertheless, such predictions are derived from models that

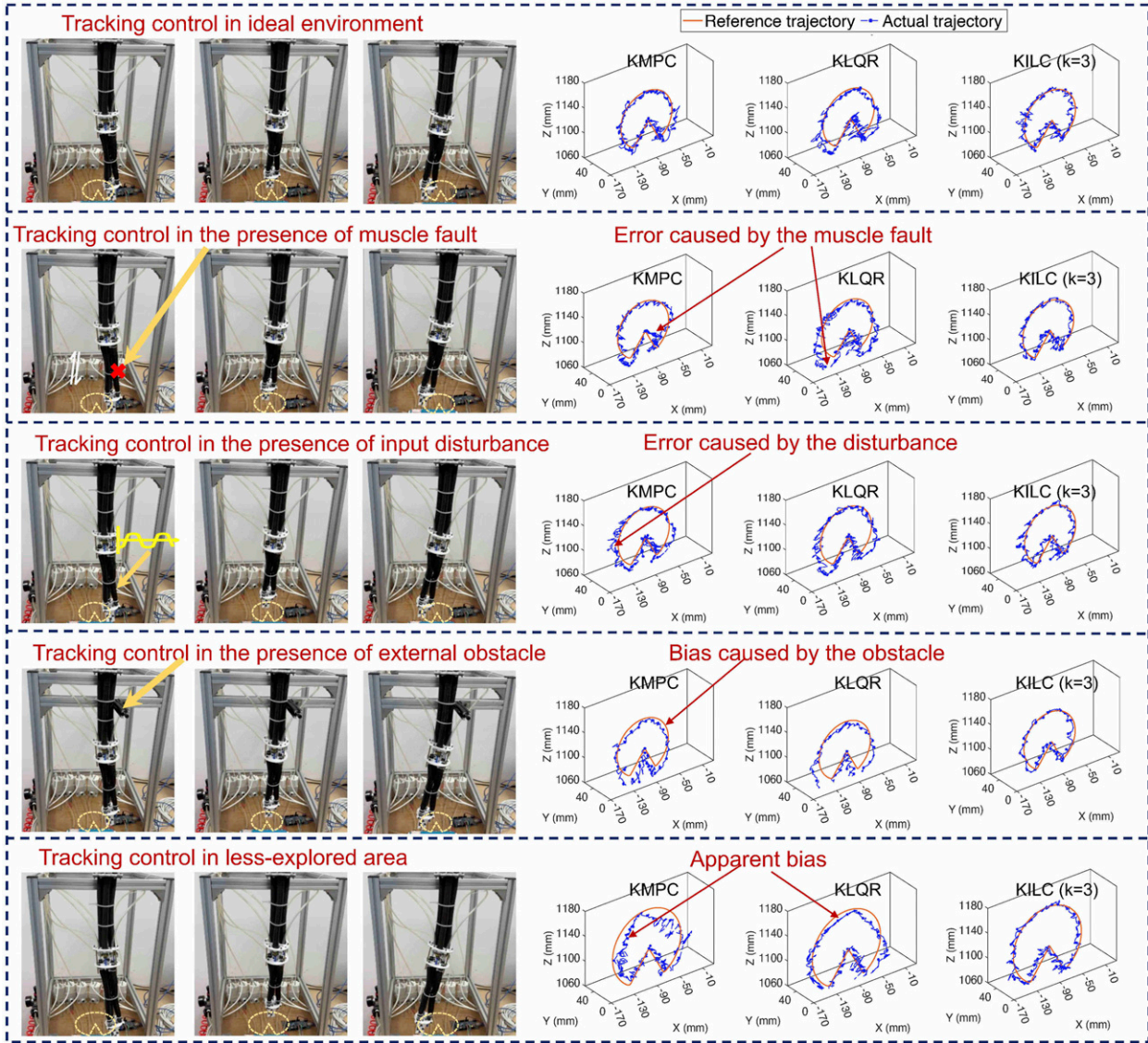


Figure 10. Specific experiment results on the pneumatic continuum robot, including snapshots of the robot (controlled by our KILC method in the third iteration) and trajectory comparisons for different methods.

may introduce considerable inaccuracies, especially under non-ideal or uncertain conditions. Consequently, the performance of the KMPC method degrades in uncertain environments. Similarly, the KLQR method also has certain shortcomings. The formula for the KLQR method is shown in Haggerty et al. (2023). The feedforward mapping matrix is learned from the static configuration data of the robot as an inverse mapping that directly maps the desired state to the actuation signal. The gain matrix in the feedback component is the LQR gain, also known as the feedback gain. This gain matrix is obtained by collecting the dynamic data of the robot and solving the LQR problem based on these data. According to the results in the literature (Haggerty et al., 2023), the feedforward component occupies a significantly larger proportion of the control signal compared to the feedback component. Therefore, if uncertainties in the environment affect the robot model, the

feedforward mapping matrix learned in an ideal environment will deviate from the actual situation, leading to inappropriate actual signal. This ultimately results in poor performance of the KLQR algorithm. Additionally, the feedback gain is a fixed matrix obtained from solving the LQR problem rather than a manually tunable parameter. We attempted to manually increase the feedback gain, for example, by multiplying feedback item by 1.5 or 2. However, instead of decreasing the error, this adjustment led to divergence. Another noteworthy observation from experiments with the motor-driven continuum robot is that the KLQR method exhibited considerable errors even in the ideal environment, as shown in Figure 6. This issue also arises from deviations in feedforward component. Specifically, at the start of each control process, the actuation signal reading of the robot is zero. However, in practice, it is challenging to ensure that the robot always starts from

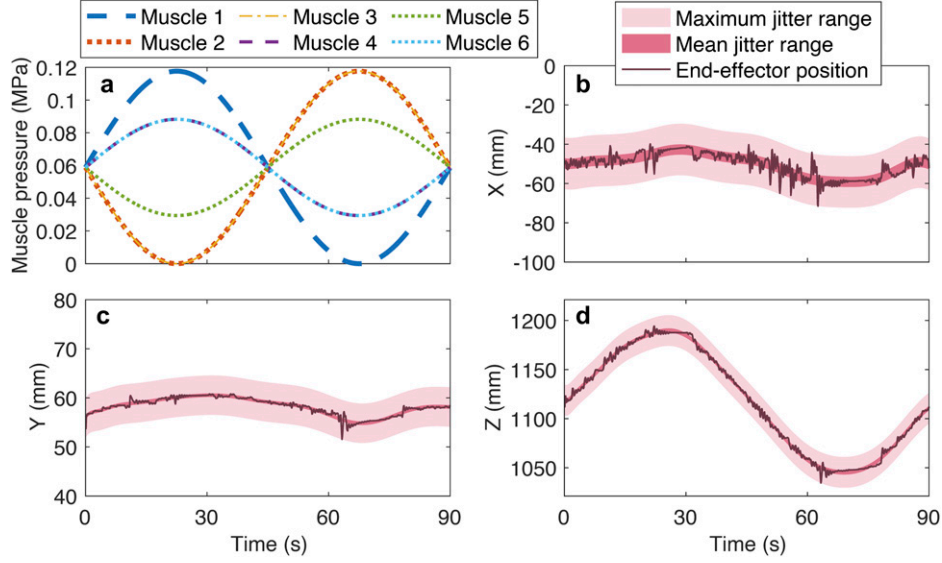


Figure 11. Stability test for the pneumatic continuum robot. (a) Smooth actuation input to the robot. (b) Tip position along the X -axis and its ranges. (c) Tip position along the Y -axis and its ranges. (d) Tip position along the Z -axis and its ranges.

Table 3. Tracking Errors and Time Costs Associated With Different Methods in the Tracking Task Performed by the Pneumatic Continuum Robot. The Best Result in Each Column is in Bold.

Method	RMSE (mm)					Time cost (ms)
	Ideal environment	Muscle fault	Input disturbance	External disturbance	Less-explored area	
KMPC	6.45	8.55	8.64	12.91	19.15	8.10
KLQR	8.90	13.43	11.02	14.27	10.88	0.30
KILC ($k = 1$)	6.76	6.78	6.92	6.77	7.26	0.36
KILC ($k = 3$)	6.11	6.05	6.41	6.15	6.89	0.36

exactly the same position. Consequently, the zero state during data collection may correspond to a different task-space state (i.e., end-effector position) than the zero state during control, resulting in significant tracking errors even without external disturbances. In contrast, the pneumatic continuum robot was less affected by this issue. Due to its structure and actuation mechanism, its zero state always remains nearly unchanged.

5. Discussion and conclusion

To address the challenges arising from the structural complexity and material softness of continuum robots in analytical modeling and precise control, data-driven control methods have gained widespread adoption. As conventional data-driven control methods do not establish explicit models, Koopman-operator-based approaches have been proposed to develop explicit models for continuum robots in a data-driven manner, thus enabling the application of mature control techniques such as MPC (Bruder et al., 2019, 2021; Thamo et al., 2023) and LQR (Haggerty et al., 2023). However, the convergence, robustness, and generalizability of existing Koopman-

operator-based methods remain unexplored, both theoretically and experimentally.

This work proposes learning the continuous-time model of continuum robots using a DKN, which bridges the gap between unknown robot models and model-dependent iterative learning control, and offers a novel control framework for continuum robots. Theoretically, we have proven that the proposed KILC method ensures the convergence of trajectory tracking tasks in the presence of various uncertainties and disturbances through iterative learning. Experimental results indicated that the proposed KILC method precisely fulfilled the tracking control tasks of continuum robots and demonstrated strong generalizability and robustness across various scenarios. Additionally, the performance of our KILC method can be further improved through iterative learning. Furthermore, an important finding is that our KILC method is computationally efficient.

The comparative studies have revealed several key findings. First, both the KLQR method and our KILC method demonstrated greater computational efficiency compared to the KMPC method, a factor crucial for the real-time performance of continuum robots. Second, as

illustrated in Figure 6(c) and Figure 9(c), our proposed KILC method outperforms the KMPC and KLQR methods across most metrics, despite their comparable performance in computational efficiency or accuracy. The KMPC and KLQR methods are highly susceptible to actuation faults and disturbances, whereas our proposed KILC method is significantly more robust. Third, an unexpected finding is that the performance of the KMPC and KLQR methods varies slightly depending on the type of robot. For example, the KMPC method exhibited poor performance in disturbance rejection when controlling the motor-driven continuum robot but performed better with the pneumatic continuum robot. In contrast, the performance of our KILC method remains consistent across different robots. Besides, our KILC method also performed well in less-explored areas while the compared methods exhibited poorer performance. These results demonstrate that our proposed method possesses greater generalizability than the KMPC and KLQR methods. In summary, the above results underscore the advantages of our KILC method in terms of accuracy, robustness, generalizability, and efficiency.

While the proposed KILC method can achieve satisfactory results in the initial iteration, additional iterations are necessary to effectively deal with various scenarios. Consequently, the primary limitation of this work is that the full potential of this method can only be realized in repetitive tasks. Despite being experimentally validated using two specific continuum robots, the proposed KILC method has the potential to control other morphologically different continuum robots or even different types of robotic systems (Chen et al., 2024; Mamakoukas et al., 2021; Ren et al., 2023), which presents an intriguing avenue for further research. In our KILC method, high feedback gains were adopted. Due to their softness and elasticity, continuum robots can passively adapt to the environment. Since high gains do not significantly affect the softness of robot body, we do not need to consider the softness preservation problem (Della Santina et al., 2017; Pierallini et al., 2022, 2023a). Only when controlling some special robots, for example, variable-stiffness robots (Bruder et al., 2023), it may be necessary to consider the problem of softness preservation, which could be achieved by employing mostly feedforward control action with low feedback gains. This is also a direction worth studying in the future.

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Supplemental Material

Supplemental material for this article is available online.

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